DE¬8307 **11**

DISTANCE EDUCATION

M.C.A.(S) DEGREE EXAMINATION, MAY 2014.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the six elements of a computers and Communications Technology. (10)

   (b) Discuss about developments in Computer Technology and Communications Technology. (10)

2. (a) Write in detail about software for cyberspace. (10)

   (b) Describe presentation graphics software in detail. (10)

3. (a) Write about desktop publishing, Integrated software. (10)

   (b) Explain hard disks and diskettes in detail with diagram. (10)

4. (a) What are the characteristics of video conferencing and picture phones. (10)

   (b) Explain the practical uses of communications and connectivity. (10)

5. (a) How do analog and digital signals and modems work, and what are alternatives to modems. (10)

   (b) What are the types, features and advantages of networks. (10)

6. (a) Explain the five generations of programming languages. (10)

   (b) Discuss about database management systems and its advantages and disadvantages. (10)

7. (a) Explain the four types of database organization and how do they work? (10)

   (b) Explain some third generation languages and it's applications. (10)

8. (a) What are the six phases of the system development life cycle. (10)

   (b) How does OOP work, and explain the OOP based languages. (10)

DE–8308 **12**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) (i) Convert $(11001101 \cdot 11001100)_2$ into hexadecimal and octal equivalents. (4)

(ii) How will you convert a decimal number into a binary number? (2)

(iii) What do you mean by Alphanumeric codes? (4)

(b) (i) How will you find the 1's and 2's complement of a binary number? Give two examples of each. (4)

(ii) Compare Boolean algebra with arithmetic and ordinary algebra. (6)

2. (a) (i) Represent the decimal number 8620 in BCD and excess-3 code. (4)

(ii) State and prove De Morgan's theorems. (6)

(b) Write notes on Number codes and Character codes. (10)

3. (a) Convert the following expression to sum-of-products form : $(A+B)(\overline{B}+C)(\overline{A}+C)$. (10)

   (b) Convert the following expression to product-of-sums form : $(A+\overline{B}+C)(AB+\overline{A}C)$. (10)

4. (a) Draw Karnaugh map for the expression $XY\overline{Z}+\overline{X}\overline{Y}Z$. (10)

   (b) Write notes on Encoder and Decoder. (10)

5. (a) Explain the functions of parallel binary adder with the logic diagram. (10)

   (b) Discuss the steps involved in the design of an Arithmetic logic unit. (10)

6. (a) Explain the functions of Master-Slave flip-flop with its logic diagram. (10)

   (b) What do you mean by the following memory elements? (10)

      (i) RAM

      (ii) ROM

      (iii) Magnetic Disk

      (iv) Magnetic Tape.

7. (a) Explain the workings of a BCD counter. (10)

   (b) Describe the functions of Shift Registers. (10)

8. (a) Discuss about Instruction cycle and Execution cycle. (10)

   (b) Write about the functions of DMA in detail. (10)

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DE–8309 **13**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2014.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the different data types in C. (10)

   (b) Write a note on gets and puts function. (10)

2. (a) What is a function? How can arguments be passed to a function in C? (10)

   (b) Write a note on multi dimensional arrays. (10)

3. (a) What is a pointer? Write a program in C to prove the use of array of pointers. (10)

   (b) Write a program in C to open and close a data file. (10)

4. How can structures be defined? Explain with a help of a program in 'C'. (20)

5. Explain the use of stacks. Write procedures for stack functioning. (20)

6. (a) What is a linked list? How are they represented? (10)

   (b) Write a note on trees. (10)

7. (a) What is a doubly linked list? Explain with diagrams. (10)

   (b) What are the applications of trees? (10)

8. (a) What is a macro? Explain with a 'C' program. (10)

   (b) Write a note on strings and its functions. (10)

DE–8310 **14**

DISTANCE EDUCATION

M.C.A (S) DEGREE EXAMINATION, MAY 2014.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain how to work with multiple windows.

   (b) Explain various menu commands in MS-Office.

2. (a) Explain about cut, copy, paste in the word document.

   (b) Explain about spell checking using MS-Word.

3. (a) Explain mail-merge in MS-word.

   (b) How to insert date and time in MS-word?

4. (a) How to format rows and columns using MS-Excel?

   (b) Explain how to create chart using MS-Excel.

5. (a) Explain about searching and sorting the excel database.

   (b) How to resize rows and columns in Excel?

6. (a) Explain create, edit, add or delete a slide using power point.

   (b) How to add speaker notes using power point?

7. (a) How to create organization chart using power point?

   (b) Explain how to add, edit and delete fields using MS-Access.

8. (a) How to set a primary key using MS-Access?

   (b) Create and save a query using MS-Access.

DE–8311 **21**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) (i) Discuss in detail about object oriented programming paradigm. (5)

   (ii) What are the basic concepts of OOPS? (5)

   (b) (i) Compare C and C++. (5)

   (ii) What are the fundamentals of C++? (5)

2. (a) Discuss the different types of array with suitable example. (10)

   (b) Explain the control structures with example. (10)

3. (a) (i) What is an iterative statement? (3)

   (ii) How does a C++ structure differ from C++ class? Discuss. (7)

   (b) What is argument passing? Discuss pass-by-value with examples. (10)

4. (a) Discuss the following with examples.

   (i) Scope of the variables (5)

   (ii) Default arguments. (5)

   (b) What is Macros and inline function? Explain with example. (10)

5. (a) Discuss about constructor and destructor in C++ with example. (10)

   (b) What is inheritance? Describe in detail about Multiple and hybrid inheritance with example. (10)

6. (a) Describe the following :

   (i) Type conversion (5)

   (ii) Operator overloading. (5)

   (b) Discuss about the virtual function and polymorphism with suitable example. (10)

7. (a) Write a C++ program to read a file and write into a another file. (10)

   (b) Discuss in detail about file pointer with example. (10)

8. (a) List the stream classes for console operation. Give example. (10)

   (b) Describe the following :

   (i) End - of - file (5)

   (ii) File direction. (5)

DE–8312 **22**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are all the key elements of a system? Explain. (10)

   (b) Discuss the characteristics of a system. (10)

2. (a) What is the role of system analyst? Explain. (10)

   (b) Discuss the interpersonal skills of a system analyst. (10)

3. (a) Explain any four fact finding techniques. (10)

   (b) Write short notes on DFD. Give an example. (10)

4. (a) Write a note on structured design technique. (10)

   (b) Discuss the role of cost and benefits analysis in developing a system. (10)

5. (a) Explain the various form design. (10)

   (b) Discuss the significance of input design for data entry. (10)

6. (a) Explain the methods of file organization. (10)

   (b) Explain the advantages of relational databases. (10)

7. Write short notes on the following :

   (a) Break even analysis

   (b) Physical design

   (c) System flow chart. (6 + 6 + 8)

8. (a) Explain the steps involved in system maintenance. (10)

   (b) Discuss the various steps involved in implementation. (10)

DE–8313 **23**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What is Event Procedures? Explain with an example? (10)

   (b) Draw the Toolbox controls. Explain each controls. (10)

2. (a) Explain various Data types in VB. (10)

   (b) Explain the properties of

      (i) Command Button

      (ii) Option Button. (10)

3. (a) What is Input Box function? Demonstrate with an example program. (10)

   (b) How will you add and Remove control in a VB form? Discuss with standard VB Controls. (10)

4. (a) Discuss the various string functions with an example coding. (10)

   (b) Explain the different types of arrays. (10)

5. (a) Discuss the properties and methods of List and Combo boxes. (10)

   (b) Define procedures. Discuss its argument passing mechanism with an example. (10)

6. (a) Discuss in detail about Menu Editor. (10)

   (b) Discuss briefly about global procedures and variables. (10)

7. (a) How do you implement animation in a VB project? Explain with an example program. (10)

   (b) Explain file system controls. (10)

8. (a) Explain sequential file processing. Compare sequential and Random File Access. (10)

   (b) Explain in detail about ADO Data Control with an example. (10)

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DE–8314 **24**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

Second Semester

Computer Applications

COMPUTER GRAPHICS

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain Bresenham's line drawing algorithm in detail. (10)

   (b) Describe the applications of computer Graphics. (10)

2. (a) Explain the various hard copy output devices in detail. (10)

   (b) Explain circle generation algorithm in detail. (10)

3. (a) Explain the basic two dimensional transformations with a neat diagram. (10)

   (b) Explain line segment and point clipping in detail. (10)

4. (a) Explain the Sutherland Hodgeman algorithm in detail. (10)

   (b) Differentiate window and viewport with a diagram. (10)

5. (a) Explain the three dimensional transformations of translation and scaling in detail. (10)

   (b) Explain the matrix representations of basic three dimensional transformations and composite transformations in detail. (10)

6. (a) Describe the projection plane and view volume. (10)

   (b) Explain Cohen Sutherland algorithm in detail. (10)

7. (a) Explain the concept of viewing process. (10)

   (b) Explain Z-buffer algorithm in detail. (10)

8. (a) Explain the user's model in detail. (10)

   (b) Explain the information display in detail. (10)

DE–8315 **31**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $\neg(P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q) \Leftrightarrow (\neg P \vee Q)$. (10)

   (b) Obtain the principal conjuctive normal form of the formula S given by $(\neg P \rightarrow R) \wedge (Q \rightleftarrows P)$. (10)

2. (a) Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$.

   (b) If the universe of discourse is the set $\{a,b,c\}$, eliminate the quantifiers in the formula $(x)R(x) \wedge (x)S(x)$. (10)

3. (a) Show that $A \subseteq B \Leftrightarrow A \cup B = B$. (10)

   (b) If relations R and S are reflexive, symmetric and transitive, show that $R \cap S$ is also reflexive, symmetric and transitive. (10)

4. (a) Given a set $S = \{1,2,3,4,5\}$ find the equivalence relation on S which generates the partition $\{\overline{1,2}, \overline{3}, \overline{4,5}\}$. (10)

(b) If S denote the set of all the partial ordering relations on a set P. Define a partial ordering relation on S and interpret this relation in terms of the elements of P. (10)

5. (a) Show that $x * y = x^y$ is a binary operation on the set of positive integers. (10)

(b) Show that $A \cap (B \cup C) = (A \cap B) \cup (a \cap C)$. (10)

6. (a) If $< M, * >$ be a monoid. Then prove that there exists a subset $T \subseteq M^M$ such that $< M, * >$ is isomorphic to the monoid $< T, \circ >$. (10)

(b) Show that every finite semigroup has an idem potent. (10)

7. (a) Prove that, every finite group of order n is isomorphic to a permutation group of degree n. (10)

(b) If $< G, * >$ and $< H, \Delta >$ be groups and $g : G \to H$ be a homomorphism. Then prove that the Kernel of g is a normal subgroup. (10)

8. (a) Prove that in a simple digraph, the length of any elementary path is less than or equal to $n-1$, where n is the number of nodes in the graph. (10)

(b) Show by means of an example that a simple digraph in which exactly one node has indegree 0 and every other node has indegree 1 is not necessarily a directed tree. (10)

DE–8316 **32**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the operating system structure in detail.(10)

   (b) Describe the concept of system calls in detail. (10)

2. (a) Discuss about the mutual exclusion in detail with examples. (10)

   (b) Write in detail about the monitors with diagram.(10)

3. (a) Explain the round robin scheduling and multiple queue scheduling in CPU scheduling in detail with diagrams. (10)

   (b) Describe the priority scheduling and shortest job scheduling in CPU scheduling in detail with diagrams. (10)

4. (a) Explain the I/O devices and device controllers in detail. (10)

   (b) Describe the device drivers and interrupt handlers in detail. (10)

5. (a) Write in detail about deadlock avoidance algorithm with diagrams. (10)

   (b) Explain the deadlock detection and recovery in detail. (10)

6. (a) Explain the page replacement algorithms in detail. (10)

   (b) Describe the concept of segmentation in detail with diagrams. (10)

7. (a) Write in detail about the multiprogramming with fixed partitions in detail with diagrams. (10)

   (b) Describe the multiprogramming with variable partitions in detail with diagrams. (10)

8. (a) Explain the directory structure in detail with diagrams. (10)

   (b) Discuss about the concept of disk space management. (10)

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DE–8317 **33**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

RDBMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about schema and sub schema with examples. (10)

   (b) Describe the features of DBMS. (10)

2. (a) Explain about Data manipulation facilities. (10)

   (b) Discuss about Data control facilities in Relational model. (10)

3. (a) Discuss the problems of Relational model. (10)

   (b) Explain the basic concepts in the Relational model. (10)

4. (a) Explain Client/Server systems with the diagram. (10)

   (b) Describe the benefits of ORACLE. (10)

5. (a) Explain the overview of ORACLE architecture. (10)

   (b) Explain the properties of Relational database. (10)

6. (a) Discuss about DML commands. (10)

   (b) Explain about SELECT statement with examples. (10)

7. (a) Discuss the elements of SQL language. (10)

   (b) Write the built in functions available in SQL. (10)

8. (a) Write short notes about embedded SQL statements. (10)

   (b) Create a table in ORACLE. Assume your own data. (10)

DE–8318 **34**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about file and directories in Unix with examples. (10)

   (b) Write in detail about grep and filters in Unix with examples. (10)

2. (a) Describe the command line structure and meta characters in shell in detail with example. (10)

   (b) Explain any two loop statements in shell programming with example programs. (10)

3. (a) Describe the various conditional expressions statements in shell programming with example programs. (10)

   (b) Discuss about executing command without invoking a new process with example shell programs. (10)

4. (a) Discuss about trap and catching interrupts with example shell programs. (10)

   (b) Write in detail about pick and news command with example shell programs. (10)

5. (a) Describe the standard input and output in Unix in detail with examples. (10)

   (b) Explain the Unix system calls in detail with examples. (10)

6. (a) Discuss about on bug and debugging with examples in Unix. (10)

   (b) Write in detail about program arguments and file access in Unix with examples. (10)

7. (a) Discuss about a screen at a time printer in Unix.(10)

   (b) Write in detail about inodes and processes in Unix with examples. (10)

8. (a) Write a shell program for four function calculator (Addition, subtraction, multiplication and division). (10)

   (b) Explain the functions and procedures in detail in Unix shell with examples. (10)

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DE–8319 **35**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Explain the various reading styles.
2. Discuss the process of article writing.
3. What are the different listening tests?
4. Write an essay on the types of interviews.
5. How could one's personality be tested through group discussion?
6. How will you manage the process of negotiation?
7. Comment on the techniques for delivering presentation.
8. Write short notes on :
   (a) Downward communication
   (b) Seminar
   (c) Body language
   (d) Faster reading.

DE–8320 **41**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the following :

   (i) user of computer network. (5)

   (ii) network hardware and network software. (5)

   (b) Describe the telephone system in detail with diagram. (10)

2. (a) Explain how twisted pair can be used as transmission media. (12)

   (b) Discuss about the asynchronous transfer mode. (8)

3. (a) Explain the error detection and correction codes in detail. (10)

   (b) Discuss the physical layer transmission media with suitable diagrams. (10)

4. (a) Discuss about the concept of petri net models. (10)

   (b) Discuss about ICMP. (10)

5. (a) Explain in detail about RARP. (10)

   (b) Explain in detail about any three network routing algorithms with diagram. (10)

6. (a) Explain in detail about TCP. (13)

   (b) Explain in detail about UDP. (7)

7. (a) Discuss about the flow control and buffering. (10)

   (b) Explain protocols for Gigabit network. (10)

8. (a) Explain in detail about secret and public key algorithms. (10)

   (b) Discuss about the electronic mail privacy. (10)

DE–8321 **42**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. The following transactions occur in the purchase and issue of materials. Prepare a stores ledger by adopting FIFO, METHOD of issuing materials.

January 2	Purchased 4,000 units at Rs.4.00 per unit
January 20	Purchased 500 units at Rs. 5.00 per unit
February 5	Issued 2,000 units
February 10	Purchased 6,000 units at Rs. 6.00 per unit
February 12	Issued 4,000 units
March 2	Issued 1,000 units
March 5	Issued 2,000 units
March15	Purchased 4,500 units at Rs. 5.50 per unit
March 20	Issued 3,000 units

2. From the following trial balance extracted from the books of Mr. Ramesh prepare his trading, profit & loss account for the year ending 30th June 2001 and a balance sheet on that date:

Particulars	Dr. Rs.	Cr. Rs.
Cash	540	
Cash at Bank	2,630	
Purchases	40,675	
Sales		98,780
Returns inward	680	
Returns outward		500
Wages	10,480	
Fuel and Power	4,730	
Carriage on sales	3,200	
Carriage on purchases	2,040	
Stock	5,760	
Buildings	30,000	
Freeholds land	10,000	
Machinery a/c	20,000	
Patents a/c	7,500	
Salaries	15,000	
General expenses	3,000	
Insurance account	600	
Drawings account	5,245	
Capital account		71,000
Sundry debtors	14,500	
Sundry creditors		6,300
	1,76,580	1,76,580

Additional information:

(a) Closing stock was valued at Rs. 6,800

(b) Machinery is to be depreciated at the rate of 10% and patents at the rate of 20%

(c) Salaries for the month of June 1965 amounting to Rs. 1,500 were unpaid.

(d) Insurance includes a premium of Rs. 170 on a policy, expiring on 31st December 1965.

(e) Wages include a sum of Rs, 2000 spent on the erection of a cycle shed for employees and customers.

(f) Create reserve for doubtful debts at 5% on sundry debtors.

3. Following are the comparative balance sheets of a company:

Liabilities	1989	1990	Assets	1989	1990
Share capital	70,000	74,000	Cash	9,000	7,800
Debentures	12,000	6,000	Debtors	14,900	17,700
Creditors	10,360	11,840	Stock	49,200	42,700
P & L a/c	10,740	11,360	Goodwill	10,000	5,000
			Land	20,000	30,000
	1,03,100	1,03,200		1,03,100	1,03,200

Additional information:

(a) Dividends were paid totaling Rs. 4,000

(b) Land was purchased for Rs. 15,000.

You are required to prepare a statement showing the changes in working capital and fund flow statement.

4. From the following figures, you are required to calculate:

   (a) Fixed Expenses

   (b) P/V Ratio

   (c) Break Even Sales

   (d) Sales to earn a profit of Rs. 12,000.

	Sales	Profit
Period- I	1,00,000	15,000
Period – II	1,20,000	23,000

5. What are the various applications of the Break-even Charts? Enumerate the various criticisms usually leveled against Break-even charts.

6. What is Overheads? Explain the classification of overhead.

7. Draw up a flexible budget for overhead expenses on the basis of the following data and determine the overhead rates at 70%, 80% and 90% plant capacity.

	Capacity		
	At 70%	At 80%	At 90%
		In Rs.	
Variable overheads:			
Indirect labour	-	12,000	-
Stores including spares	-	4,000	-
Semi-variable overheads:			
Power (60% fixed, 40% variable)	-	20,000	-

	Capacity		
	At 70%	At 80%	At 90%
		In Rs.	
Repairs and maintenance:			
(60% fixed, 40% variable)	-	2,000	-
Fixed overheads:			
Depreciation	-	11,000	-
Insurance	-	3,000	-
Salaries	-	10,000	-
Estimated Direct Labour hours (1,24,000 hours)			

8. The following data represents the ratios pertaining to Empson Co Ltd ., for the year ending 31 March 1999.

	Rs.
Annual sales	40,00,000
Sales to Net worth	4 times
Current liabilities to worth	50%
Total debts to net worth	80%
Current ratio	2.2 times
Sales to inventory	8 times
Average collection period	40 days
Fixed assets to Net worth	70 days

From the above particulars, prepare the balance sheet of Empson Co Ltd. as on 31 March 1999. Assume that all sales are made on credit.

DE–8322 **43**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) A company makes two kinds of leather belts. Belt A is high quality belt, and belt B is of lower quality. The respective profits are Rs. 4.00 and Rs. 3.00 per belt. Each belt of type A requires twice as much time as a belt of type B, and if all belts are of type B, the company could make 1000 per day. The supply of leather is sufficient for only 800 belts per day (Both A and B are combined) Belt A requires a fancy buckle and only 400 per day are available. There are 700 buckles a day available for belt B. Determine the optimal product mix by using graphical method. (10)

(b) Use simplex method to solve the following. (10)

Linear programming problem.

Maximize $Z = 4x_1 + 10x_2$

Subject to the constraints :

$2x_1 + x_2 \leq 50$

$2x_1 + 5x_2 \leq 100$

$2x_1 + 3x_2 \leq 90$

$x_1 \geq 0$ and $x_2 \geq 0$

2. (a) Find the starting solution in the following transportation problem by Vogel's approximation method. Also find the optimum solution. (10)

	D_1	D_2	D_3	D_4	Supply
S_1	3	7	6	4	5
S_2	2	4	3	2	2
S_3	4	3	8	5	3
Demand	3	3	2	2	

(b) A departmental head has 4 subordinates and four tasks to be performed. The subordinates differ in efficiency, and the tasks differ in their intrinsic difficulty. His estimate of the time each man would take to perform each task is given in the matrix below.

Men

Task	E	F	G	H
A	18	26	17	11
B	13	28	14	26
C	38	19	18	15
D	19	26	24	10

How should the tasks be allocated, one to a man, so as to minimize the total man hours. (10)

3. (a) A T.V. Repairman finds that the time spent on his jobs has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which they came in and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day. What is repairman's expected idle time each day? How many jobs are ahead of the average set just brought in? (5)

(b) At a certain petrol pump, customers arrive in a Poisson process with an average time of 5 minutes between arrivals. The time interval between servers at the petrol pump follows an exponential distribution and the mean time taken to service is 2 mts. Find the following

(i) Expected average queue length

(ii) Average number of customers in the system

(iii) Average time a customer has to wait in the queue

(iv) Average time a customer has to spend in the system

(v) By how much time the flow of the customer be increased to justify the opening of another service point where the customer has to wait for 5 minutes for the service? (15)

4. At a telephone Booth, suppose that the customers arrive with an average time of 1.2 time units between one arrival and the next. Service times are assumed to be 2.8 time units. Simulate the system for 12 time units by assuming that the system starts at t = 0. What is the average waiting time per customer? (20)

5. A project consists of a series of tasks labelled $A, B, \ldots. H, I$ with the following relationships ($W < X, Y$ means X and Y cannot start until W is completed, $X, Y < W$ means W cannot start until both X and Y are completed) with this notation construct the network diagram using the below constraints.

$A < D, E; B, D < F; C < G; B, G < H; F, G < I$. Find also minimum time of completion of the project when the time (in days) of completion of each task is as follows : (20)

Task:	A	B	C	D	E	F	G	H	I
Time:	23	8	20	16	24	18	19	4	10

6. (a) Define the following terms :

   (i) Pessimistic time

   (ii) Optimistic time

   (iii) Most likely time. (6)

   (b) What is critical path? Write the differences between PERT and CPM. (14)

7. (a) Derive EOQ formula for the manufacturing model without shortages. (10)

   (b) A manufacturing company purchases 9,000 parts of a machine for its annual requirements ordering one month usage at a time. Each part costs Rs. 20. The ordering cost per order is Rs. 15 and the carrying charges are 15% of the average inventory per year. You have been assigned to suggest a more economical purchasing policy for the company. What advice would you offer and how much would it save the company per year? (10)

8. (a) Machine A costs Rs. 9,000. Annual operating costs are Rs. 200 for the first year and then increase by Rs. 2,000 every year. Determine the best age at which to replace the machine. Find the average yearly cost of owning and operating the machine?

   (b) Machine B costs Rs. 10,000. Annual operating costs are Rs. 400 for first year and then increase by Rs. 800 every year. You now have a machine of type A which is one year old. Should you replace it with B, if so when? (20)

DE–8323 **44**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the applying object model concept in detail. (10)

   (b) Describe the evolution of the object model in detail. (10)

2. (a) Write in detail about the nature of an object. (10)

   (b) Discuss about the relationships among objects. (10)

3. (a) Explain the identifying proper classes and objects in detail. (10)

   (b) Describe the interplay of classes and objects in detail. (10)

4. (a) Write in detail about on building quality classes and objects. (10)

   (b) Discuss about the relationship among classes with examples. (10)

5. (a) Describe the overview of the view process in detail. (10)

   (b) Discuss about the Iteractive development in detail. (10)

6. (a) Discuss about the meta model. (10)

   (b) Explain the concept of patterns in detail. (10)

7. (a) Describe the concept of essentials of class diagram in detail. (10)

   (b) Write in detail about the class diagram with examples. (10)

8. (a) Explain the state diagram in detail with diagrams. (10)

   (b) Describe the deployment diagram in detail with examples. (10)

DE–8324 **45**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(Lateral) DEGREE EXAMINATION,
MAY 2014.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the issues in WWW. design.

   (b) Write a brief note on Perl.

2. (a) Give a brief note on developing intranet applications.

   (b) Discuss the various built-in functions of Javascript with an example program.

3. (a) Write a brief note on the intrinsic controls of VB script.

   (b) Discuss briefly on HTML.

4. (a) Write a brief note on SGML.

   (b) How do you create an ActiveX control to activate a web page? Explain.

5. (a) Write short notes on VDO live technology.

   (b) How do you create a custom integrated application with multiple protocols? Explain.

6. (a) Write a brief note on Images and Hyperlinks.

   (b) How do you create an ActiveX control to activate a web page? Discuss.

7. (a) Discuss briefly about how to add graphics to web pages.

   (b) What are dynamic graphics? Discuss.

8. (a) Explain briefly about Java in windows.

   (b) Write short notes on animation with a suitable example.

DE–8325 **51**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Describe the components of image processing system. (10)

   (b) Discuss the applications of image processing system. (10)

2. (a) Write in detail about sampling and quantization with diagrams. (10)

   (b) Explain in detail about the elements of digital geomentric. (10)

3. (a) Describe about edge preserving smoothing. (10)

   (b) Explain the prestoration by homomorphic filtering in detail. (10)

4. (a) Discuss about the least square error restoration. (10)

   (b) Describe minimum mean-square error restoration. (10)

5. (a) Write in detail about lossy compression method. (10)

   (b) Write in detail about Huffman coding. (10)

6. (a) Write in detail about stereo imaging algorithm. (10)

   (b) Discuss the details about processing of medical images. (10)

7. (a) Write about the optimum threshold in image processing. (10)

   (b) Discuss about multi-level thresholding in detail. (10)

8. (a) Explain the corner detection in detail. (10)

   (b) Explain the fuzzy mathematical recognition in detail. (10)

DE–8326 **52**

DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain about the quality and productive factors. (10)

   (b) Discuss about the process and project metrics. (10)

2. (a) Describe the details about role of management in software development. (10)

   (b) Explain about the software cost estimation. (10)

3. (a) Discuss about the cost factors. (10)

   (b) Describe about the planning an organizational structure. (10)

4. (a) Discuss about the defining task and scheduling a plane. (10)

   (b) Discuss about quality standards. (10)

5. (a) Describe the software risks and identification. (10)

   (b) Explain in detail monitoring and management in Software Engineering. (10)

6. (a) Discuss about software requirement specification. (10)

   (b) Explain the structure information and prototyping. (10)

7. (a) Discuss about the characteristics of specification analysis. (10)

   (b) Discuss about model level concepts. (10)

8. (a) Discuss about transform mapping design with diagrams. (10)

   (b) Explain the black box testing and control structure testing in detail with diagram. (10)

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DE–8327 **53**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Fifth Semester

VISUAL PROGRAMMING — II

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss in detail about file creation using Appwizard with example. (10)

   (b) Explain in detail about Anatomy of C++ windows programs. (10)

2. (a) Write in detail about Document object and view object. (10)

   (b) Discuss about mainframe object and APP object. (10)

3. (a) Explain in detail about mouse events. (10)

   (b) Describe the concept of adding menus to a program in detail. (10)

4. (a) How menu items are added? Explain in detail. (10)

   (b) Describe the concept of adding short cut keys in detail with suitable example. (10)

5. (a) Explain message boxes in detail. (10)

   (b) Discuss in detail about the member variables and functions to update data. (10)

6. (a) Write in detail about List Boxes and Combo Boxes. (10)

   (b) Write a VC++ menu driven program to draw geometric shapes. (10)

7. (a) Discuss in detail about sequential Access file with example. (10)

   (b) Explain MDI programs in detail. (10)

8. (a) Explain in detail about the concept of Exception Handling with example. (10)

   (b) Describe about the VC++ ODBC classes. (10)

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DE–8328 **54**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(Lateral) DEGREE EXAMINATION,
MAY 2014.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the compiler and translators. (10)

   (b) Explain the role of syntax analysis in detail. (10)

2. (a) Explain the role of Lexical analysis in detail. (10)

   (b) Explain the implementation of Lexical analyser in detail. (10)

3. (a) Consider the ambiguous grammar.

   $S \rightarrow AS / b$

   $A \rightarrow SA / a$ (10)

   Construct the parsing table using SLR table construction algorithm.

   (b) Explain derivation and parse trees in detail. (10)

4. (a) Explain the various steps involved in constructing canonical LR parsing table in detail. (10)

   (b) Explain TOP down parsing in detail. (10)

5. (a) Explain syntax-direct translation in detail. (10)

   (b) Write in detail about postfix notations and postfix translations with example. (10)

6. (a) Describe the contents of a symbol table in detail. (10)

   (b) Explain error detection and error recovery in detail. (10)

7. (a) Explain the representation of scope information in detail. (10)

   (b) Explain syntactic and semantic errors. (10)

8. (a) Explain the representation of basic blocks in detail. (10)

   (b) Explain simple code generator in detail. (10)

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DE–8329 **55**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write notes on information and production factor. (10)

   (b) Describe the Data mining activities in marketing. (10)

2. (a) Describe preliminary analysis of the data set using traditional query tools. (10)

   (b) Discuss on data selection and data cleaning. (10)

3. (a) Illustrate Association rules through examples. (10)

   (b) Discuss on knowledge discovery in databases environment. (10)

4. (a) With a neat sketch, explain Data Warehouse architecture. (10)

   (b) Discuss on partitioning strategy. (10)

5. (a) Describe database schema. (10)

   (b) Write notes on aggregations and data marting. (10)

6. (a) Explain about operational design of Data Warehouse. (10)

   (b) Discuss on issues of security in Data Warehouse. (10)

7. (a) Describe backup and recovery process in Data Warehouse. (10)

   (b) Explain service level agreement in detail. (10)

8. (a) Write notes on tuning the Data Warehouse. (10)

   (b) Bring out the features of Data Warehouse. (10)

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DE–8330 **61**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE
EXAMINATION, MAY 2014.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Explain the various types of projects and its problems. (20)

2. Explain the network planning model with a neat sketch. (20)

3. Explain the risk management and resource allocation in projects. (20)

4. How can critical activities be identified for projects? Explain. (20)

5. What is software configuration management? Explain. (20)

6. What is prototyping? Explain in detail. (20)

7. Write a note on PRINCE project management with suitable diagram. (20)

8. Explain resource allocation in projects with a neat diagram. (20)

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DE–8331 **62**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Sixth Semester

Computer Applications

MULTIMEDIA SYSTEMS

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the need and benefits of multimedia in detail. (10)

   (b) Discuss about multimedia platforms. (10)

2. (a) Explain the system components in detail. (10)

   (b) Explain the commercial tools in multimedia in detail. (10)

3. (a) Discuss about text, image and graphics. (10)

   (b) Compare analog versus digital video and explain the steps to add a sound to the multimedia project. (10)

4. (a) Briefly explain about the different techniques used for video compression. (10)

   (b) Discuss about MPEG and DVI. (10)

5. (a) Explain about JPEG standards. (10)

   (b) Explain about redundancy and visibility in detail. (10)

6. (a) Discuss about component classes in multimedia frameworks. (10)

   (b) Explain the object and classes in object oriented multimedia frameworks in detail. (10)

7. (a) Discuss about format classes in multimedia frameworks. (10)

   (b) Explain about media classes in multimedia frameworks. (10)

8. (a) Explain about the multimedia in real world. (10)

   (b) Explain the architecture and operations in multimedia environments in detail. (10)

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DE–8332 **63**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2014.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the model of an artificial neuron. (10)

   (b) Explain the neural network architecture. (10)

2. (a) What are the applications of artificial neural networks? (10)

   (b) Discuss about the supervised and unsupervised learning methods. (10)

3. (a) Explain about memory based learning. (10)

   (b) Explain the method of Boltzmann learning. (10)

4. (a) Explain the multilayer perceptron model. (10)

   (b) Discuss the back propagation network. (10)

5. (a) Write in detail any two applications of back propagation networks. (10)

   (b) Discuss the structure of counter propagation networks. (10)

6. (a) Discuss in detail Hopfield networks. (10)

   (b) Explain the energy function for bi-directional associative memory. (10)

7. (a) Discuss about the applications of associative memory. (10)

   (b) Explain the simplified ART architecture. (10)

8. (a) Explain in detail about cognitions. (10)

   (b) Discuss about optical neural network. (10)

DE–8333 **64**

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, MAY 2014.

Sixth Semester

C–SHARP (C #)

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain in detail about Microsoft .Net. (10)

   (b) Discuss in detail about the components of .net. (10)

2. (a) Describe about the automatic resource management. (10)

   (b) Discuss in detail about code reuse and security in C #. (10)

3. (a) Explain in detail about the features of C#. (10)

   (b) Write short note on the following:

      (i) VOS type systems (5)

      (ii) Jitters (5)

4. (a) Explain various data types in C#. (10)

   (b) Discuss the steps to create, compile and execute a C# program. (10)

5. (a) Explain in detail about constructors in C# with example. (10)

   (b) Explain about method definition and method calling in C# with example. (10)

6. (a) Explain the concepts of exception handling in C# with suitable example. (10)

   (b) Explain in detail about the control statements in C# with example. (10)

7. (a) How simple client created in C#? Explain with example. (10)

   (b) Discuss about the concept of name space in C# with example. (10)

8. (a) Write in detail about documentation and comments in XML with example. (10)

   (b) Discuss in detail about configuring and deployment. (10)

DE–8815 **15**

DISTANCE EDUCATION

M.C.A.(S) DEGREE EXAMINATION, MAY 2014.

First Semester

LAB I — DATA STRUCTURES USING C

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a C program to find the roots of a quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a C program to find the factorial of a given number using recursion.

---------- Cut here ----------

2. (a) Write a C program to copy contents of one file to another file.

   (b) Write a C program for matrix manipulation (Addition, subtraction and multiplication) Using function.

3. (a) Write a C program to find word is PALINDROME or NOT.

(Eg. The word MALAYALAM is PALINDROME and the word TAMIL is not a PALINDROME)

(b) Write a C program to sort a set of elements using insertion sort.

---------- Cut here ----------

4. (a) Write a C program to read 10 values to any array variable. use pointers to locate and display each value.

(b) Write a menu driven program in c to find an element using linear and binary search methods.

---------- Cut here ----------

5. (a) Write a C program to implement insert and delete operations on queue using array concept.

(b) Define selection sort. Write a C program to sort a set of element using selection sort.

---------- Cut here ----------

6. (a) Write a C program to sort 10 Nos. in ascending order with naming of variable and the value before and after sorting.

(b) Write a C program to convert a given number into words for numbers 1 to 5.

(Eg. 1 to one, 2 to two and 5 to five)

---------- Cut here ----------

7. (a) Write a C program for linked list implementation of queue operations.

(b) Write a C program to implement push and pop operations on stack.

8. (a) Write a C program to find the roots of a quadratic equation using function definition $ax^2 + bx + c = 0$.

(b) Write a C program to evaluate the given mathematical expression using stack.

DE–8816 **16**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

First Semester

LAB II – MS OFFICE AND INTERNET

(2005 – 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer any TWO questions choosing one from Section A and another from Section B

Each question carries 50 marks.

Break-up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

SECTION – A

1. Prepare an invitation for a function to be conducted in your institution. Use different text orientation and pictures to make it attractive.

---------- Cut here ----------

2. Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

3. Create a form letter that informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal mail merge it with an access database containing all the customer data. Prepare letters for customer whose due date falls in a specific range of dates.

---------- Cut here ----------

4. Create a worksheet with the following columns : salesman number, salesman name, city, product name, sales amount.

   Add three records for 5 different salesmen who have carried out sales of different products in different cities. Find the following.

   (a) City wise total sales

   (b) Salesman wise total sales

   (c) Product wise total sales

   (d) Grand total sales.

---------- Cut here ----------

5. The following were the observations made in certain experiments for the values and given the values of X.

X :	1	2	3	4	5	6	7	8	9	10
Y :	10	30	45	25	15	28	40	32	15	35

   Compute the following :

   (a) Find mean, median and standard deviation

   (b) Correlation coefficient between X and Y

   (c) Draw the bar chart and shading cells

   (d) Draw a chart in excel and paste it an word.

6. Draw LINE, BAR, PIE charts for the data given below :

ABC company limited

Year :	Sales	Cost	Profit
1991	1000	400	600

Add data for 10 years. From 1991 to 2000. Provide titles, legends, grids and data labels.

---------- Cut here ----------

7. Create employee table with the fields EMPNO, NAME, AGE, SEX, STREET, CITY, PIN, SALARY.

   (a) Add data for 20 employees

   (b) Write a query to display all the male employees whose salary is between 1000 and 5000 and living in city "CHENNAI"

   (c) Write a query to display all the female employees whose ages are in the range 50-60.

   (d) Show all the records in the table for the city, "TRICHY" by filtering.

---------- Cut here ----------

8. Create STUDENT table with the following fields REGNO, NAME, MARK1, MARK 2, MARK 3. Create ADDRESS table with fields REGNO, STREET, CITY and PIN.

   (a) Write a query to display REGNO, NAME, and total of all the three subject marks.

   (b) Write a query to display REGNO, NAME, STREET, CITY, PIN and total of all the three subject marks.

SECTION – B

9. Create HTML page that divides the screen space into three frames (one vertical and two horizontal of equal size). Display three different files in them with scrolling titles relevant to their contents.

---------- Cut here ----------

10. Create HTML page to display 10 courses and their high lighted features, conducted by an educational institution in a colorful format.

---------- Cut here ----------

11. Create HTML page that plays video and audio files.

DE–8817 **25**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

Second Semester

GRAPHICS USING C++ – COMPUTER LAB – III

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the ONE question for each candidate by LOT system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. Draw a ball perform animation on it.

---------- Cut here ----------

2. Draw any 3-D object and perform animation on it.

---------- Cut here ----------

3. Write a C++ program to draw non-linear objects arcs and curve.

4. Write a C++ program to manipulate the graphical object using SHADING technique.

---------- Cut here ----------

5. Draw a circle. Divide into four quadratic and fillup different colour for each quadrant.

---------- Cut here ----------

6. Create a base class shape, which contains the common elements of circle, triangle and rectangle, from which derive three classes called ball rect and triangle. Include two member functions set ( ) to set coordinates and draw ( ) which draws them. Write a main ( ) program to exercise this class using graphic- mode graphic functions.

---------- Cut here ----------

7. Write procedure and a C++ program to draw non-linear object circle and three concentric circle.

---------- Cut here ----------

8. Write a C++ program to draw to straight line using simple DDA technique.

---------- Cut here ----------

9. Write a C++ program to draw a straight line using simple Bresunhams line algorithm.

---------- Cut here ----------

10. Write procedure and a C++ program to draw non-linear object ellipse.

DE–8818 **26**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2014.

Second Semester

LAB IV — VISUAL BASIC

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each sub division carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a VB project to create a screen saver that displays a list of pictures with one second pause in between successive pictures.

   (b) Write a VB project that stores 10 student records with fields STU No., STU NAME, AGE, SEX and consolidated percentage of marks in an array. Display data fields in text boxes and provide command buttons to move to desired record.

2. (a) Write a VB project using control array that creates an ordinary calculator with appropriate command buttons. Include the following capabilities for the calculator : +, –, *, /, %.

(b) Write a VB project to display the availability of seats when the user selects the train name and date from the combo box and timings from another combo box.

---------- Cut here ----------

3. (a) Write a VB project for commercial bank operation using SB Account database with the following features.

(i) Deposit amount.

(ii) With draw amount.

(with minimum balance condition)

(b) Design and implement one ActiveX control for stop watch used for sports.

---------- Cut here ----------

4. (a) Create a VB application to make the text bold, italic, underline and also color the text. (using check box and option button).

(b) Design and implement and ActiveX control for Railway Clock (Hour display 1 to 24).

---------- Cut here ----------

5. (a) Write a VB project to display a message (example left/right mouse button is clicked) when the mouse button is clicked using mouse move event of the form.

(b) Design and implement one ActiveX control for monthly calendar.

6. (a) Write a VB project that receives a year number from a text box and month name from list box and displays number of days in the given month. Take care of leap years. Use lost focus event for list box.

(b) Write a VB project to display the availability of books when the user selects the author name from the combo box and book title from another combo box.

---------- Cut here ----------

7. (a) Write a VB project to calculate the amount when the user selects the class 'Type of the Train' from the combo box and also enters the number of passengers.

(b) Design and implement one ActiveX control for digital electric current usage measurement meter used in houses.

---------- Cut here ----------

8. (a) Write a VB project using AX DLL or EXE add a class module that would perform the following functions :

(i) Sum of Digits.

(ii) Whether the given number is Armstrong or not.

(b) Write a VB project for commercial bank operations using SB Account database with the following features.

(i) ADD New account.

(ii) Calculate simple interest and update balance taking average of last 6 month balance in the account.

(iii) Close account.

9. (a) Write a VB project for a blood bank that maintains a list of donors with address and their blood group provide the following reporting features.

Display all the donors (using data reports).

(i) With particular blood groups.

(ii) Male donors with particular blood group.

(iii) Female donors with particular blood group.

(b) Create an application using VB to move elements from one list to another list and add new items.

---------- Cut here ----------

10. (a) Write a VB project for a blood bank that maintains a list of donors with address and their blood group. Provide the following reporting features.

Display all donors (using data reports)

(i) In age group 20-30.

(ii) In a particular city.

(b) Write a VB project to display the availability of seats when the user selects the flight name from the combo box and date including timings from another combo box.

DE–8819 **36**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S)/(Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

LAB V — RDBMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Students has to answer ONE question by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. A News paper vendor buys newspapers on wholesale from a distributor for Rs. 4 and sell them in retail for Rs. 4.50. At the end of the day the unsold papers arc returned to the distributor for a Rs. 1 rebate per paper. Write a program using Oracle to prepare a port for the newspaper vendor in the following format with 9 Weeks data :

Week	Bought	Sold	Return	Profit	Loss
1	100	50	XX	XX.XX	XX.XX
2	75	60	XX	XX.XX	XX.XX
--	--	---	---	------	------
--	--	---	---	------	------
Total					

2. A salary statement contains Name, Basic pay. Allowances. Total Deduction

(includes IT). Gross pay, Net pay.

Dearness Allowance = 58% of Basic pay

Gross Pay = Basic pay + D.A

Deduction = P.F + LOAN AMT

Income Tax is calculated on the basis of annual income with the following conditions.

Annual salary	Income Tax
<= 1 Lakh	Nil
>1 Lakh but < =1.5 Lakh	10%
>1.5 Lakh but < = 3 Lakh	20%
>3 Lakh and above	30%

Write a program in SQL to prepare Salary report for 25 employees.

---------- Cut here ----------

3. A daily sales file contains records with the following fields :

Department number, date, item description sale price for each item, quantity, cost of each item. Write a program using Oracle to list all the input data and compute total amount of sales and profit. Show the output report in the following format :

Department number

Item description

sales price

quantity

cost

profit.

4. An examination has been conducted to a class of 55 students and the following are given as inputs :

Three test marks. Register No and Candidate Name.

Write a SQL program to do the following :

Assign a letter grade to each student based on the average score and list Register Number. Average score, Grade. The minimum pass for each subject is 50.

The grading System :

Average score	Grade
90-100	A
75-89	B
60-74	C
50-59	D
0-49	F-fail

---------- Cut here ----------

5. A hospital maintains blood donors record information in a file. The file contains the following fields :

Donor number, donor name, donor age, donor address, pin code, place of birth, blood group (A, B, AB & O).

Write a program using Oracle to print out the number, name and address of the donors for the following categories :

(a) Blood donors having AB blood group

(b) Blood donors in the age group between 16 to 25

(c) Female donors with 'A' blood group in the age between 18 and 25.

6. Write an SQL program for the canteen Maintenance with two tables namely MENU and BILL.

The MENU table contains the following fields:

ITEM, UNIT-Rate.

The following items are available in the canteen :

TEA, COFFEE and COOL-DRINKS

The Bill table contains the following fields :

EMPNO. NAME, DATE OF ISSUE, ITEM1, NO.OF TOKENS ISSUED FOR ITEM1, RATE1, ITEM2. NO.OF TOKENS ISSUED FOR ITEM2. RATE2. ITEM3. NO.OF TOKENS ISSUED FOR ITEM3. RATE3. TOTAL AMOUNT.

Note : An employee is allowed to get the token only once in a month.

RATE = UNIT-RATE*NO.OF TOKENS FOR PARTICULAR ITEM

TOTAL AMOUNT = RATE-1 + RATE-2 + RATE-3

Prepare a report in the following format :

S.NO.	EMP.NO	EMPNAME	PURCHASED RATE
1	--	--	--
2	--	--	--
3	--	--	--
			Total Amount

7. An airline reservation database contains the reservation table and personal table.

   Reservation table contains following fields :

   Flight Number, Class, Passenger Name. Seat Number

   Personal Table contains following fields:

   Passenger Name, Sex, Age. Marital status, Nationality

   Write a program in ORACLE to prepare the following reports :

   (a) List the passengers' names with seat number according to Class and flight numbers

   (b) Total number of married female passengers in a particular flight.

   (c) List out the Male passengers in the age between 18 to 25 for all flights.

---------- Cut here ----------

8. A company RX Pvt. Ltd states monthly salary to its employee. It consists of Basic salary. Allowance and deduction.

   D.A. - 43% of Basic Salary

   H.R.A. - 7% of(Basic Salary + D.A)

   Deductions:

   P.F. – subscribed by an capital

   LIC premium – payable by an employee, the salary saving Scheme.

   Loan premium – If any payable by an employee

   Create a master of 10 records with following fields:

   Employee number, name, designation. Basic Salary, HRA, DA, Bank A/c number. LIC premium number

   A transaction file contains

   Employee number. P.F.subscription. LIC premium amount. Loan recovery.

   Write a program in SQL to prepare a report with the following information:

   Serial number, Employee number, Bank A/c number. Name, Basic Salary, Total Allowances. Gross pay, Total deductions, Net pay.

9. Write a PLSQL program to prepare a Mark statement of a student using student table and Mark table. Assume your own data.

---------- Cut here ----------

10. Write a program in SQL for hospital Billing system with the following fields :

Patient no, patient name, age, Attending doctor, Patient-type(IN/OUT). consulting charge. Blood test charge. X-Ray charges. other test charges and total fee.

Prepare reports with the following conditions:

(a) The patients undergone blood test

(b) The patients who taken X-rays

(c) The patients who belong to in-patient category.

The common fields to be included in the above mentioned reports.

Patient No, Name, Age, Charges and Total Charges.

DE–8820 **37**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Third Semester

LAB VI — ALGORITHMS AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging and Execution	:	35
Result	:	10
Total	:	100

ONE question is to be given by the examiner to each candidate by lot system

1. (a) Write a C++ program to sort a list of numbers in ascending order using straight selection sort.

   (b) Write a Menu driven shell program to do the following:

   (i) List of files

   (ii) Process of users

   (iii) Current date

   (iv) Users of system

   (v) Quit to UNIX

2. (a) Write a C++ program to sort a list of numbers in ascending order using straight insertion sort.

(b) Write a shell program which accepts the name of the file from standard input media and then perform the following tests on it.

(i) File Existence

(ii) File Readable

(iii) File Writeable

(iv) Both readable and writeable

---------- Cut here ----------

3. (a) Write a C++ program to sort a list of numbers in ascending order using straight exchange sorting method.

(b) Write a shell program using three commands to take the pattern as well as input and output files names. If the pattern is found, display the message "Pattern Found" otherwise displays "Error". Also check the number of arguments is correct or not.

---------- Cut here ----------

4. (a) Write a C++ program to sort a list of numbers in ascending order using quick sort (Divide and Conquer) method.

(b) Write a shell program to periodically monitors the disk for existence of a file and then executes the program once the file has been located. (using UNTIL statement).

5. (a) Write a C++ program sort a list of numbers in ascending order using Heap sort method.

(b) Write a shell program to accept the name of a file from standard input media and then perform the following tests on it.

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List sorted and unsorted file

(iv) Quit

---------- Cut here ----------

6. (a) Write a C++ program to implement the Knight Tour problem using backtracking method.

(b) Write a menu-driven shell program to perform the following file operations: COPY, EDIT, RENAME, and DELETE.

---------- Cut here ----------

7. (a) Write a C++ program to implement the Traveling Salesman problem using Branch and Bound method.

(b) Write a menu-driven shell program to perform the following tasks:

(i) Enter the sentence in a file

(ii) Search a word

(iii) Quit.

8. (a) Write a C++ program to search a number using Linear and Binary search method.

(b) Write a Shell program to perform the following tasks in Hospital maintenance system.

(i) List patients in Blood group wise

(ii) List patients in age between 20 to 30

(iii) Exit.

---------- Cut here ----------

9. (a) Write a C++ recursive program to perform the following

(i) Factorial

(ii) Fibonacci series.

(b) Write a Shell program to prepare the electricity bill for the domestic consumers as per the following conditions:

Up to 100 units - Rs. 0.75/Unit

For next 100 units— Rs. 1 .50/unit

Above 200 units — Rs. 3.00/unit

Display the Electricity bill in the below format

Customer Number:

Customer Name:

Previous Reading:

Current Reading:

Consumed Units:

Total Amount:

---------- Cut here ----------

10. (a) Write a c++ program to implement the Towers of Hanoi of problem

(b) Write a Shell program to sum up the series.

$$\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots\ldots\ldots + \frac{1}{10!}$$

DE–8821 **46**

DISTANCE EDUCATION

M.C.A (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 calender Year)

Time : Three hours Maximum : 100 marks

One question is to be given by examiner to each candidate by LOT system

Each sub division carries 50 marks.

Break up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Write a menu driven Java program to do database functions using database connectivity facility. The functions include display, edit, insert and delete a record from a table.

(b) Using atleast 20 HTML tags, create a screen with a string “WEB design”.

2. (a) Write a VB script to create a calendar for given month and year.

(b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

3. (a) Write a Java program using applet to display any 3 images when 3 buttons in the border layout are clicked. The image should be displayed at the center.

(b) Write a Java script to create a window by using the confirm message.

---------- Cut here ----------

4. (a) Write a VB script to do the following

(i) Check the given password

(ii) Change the existing password.

(b) Write a Java program using applet to display the different color and fonts.

---------- Cut here ----------

5. (a) Write a Java script to create order form to select the house articles.

(b) Design and publish a web page for a college.

6. (a) Write a swing program to create buttons with

   (i) tool tip next

   (ii) Image

   (iii) Border

   (iv) Short cut key

   (b) Write a java program to create a color as the background.

---------- Cut here ----------

7. (a) Write a swing program to create the tabbed panels.

   (b) Create a web page in the format of front page of a newspaper using text link/ align the text with colors.

---------- Cut here ----------

8. (a) Write a VB script to do the following :

   (i) Display the current date

   (ii) Find the difference between two dates

   (iii) Find the age of a person by providing data of birth.

   (b) Develop a picture gallery having atleast 5 pages. Each of them is having several pictures with suitable information.

---------- Cut here ----------

9. (a) Write a Java script to create color palet and display the text in the color chosen from the palet with proper background color.

   (b) Write a Java program using applet to create the frames and its controls.

10. (a) Write a Java program using applet to display the dialogue menu in applet.

(b) Develop a web page for job recruitment agency in an IT industry.

DE–8822 **47**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Fourth Semester

LAB – VIII – NETWORK LAB

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any ONE choosen by the candidate through lot system.

Break-Up of Marks :

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a Java program to multiply two numbers using RMI client/server technique.

   (b) Write a Java program for passing simple message.

---------- Cut here ----------

2. (a) Write a java program to find factorial to given number using RMI client/server technique.

   (b) Write a java socket program for Echo server and Echo Client communication.

3. (a) Write a java program to find the given number is ODD or EVEN using RMI client/server technique.

   (b) Write a Java program to read a file from the remote system.

---------- Cut here ----------

4. (a) Write a java program to solve the towers of Hanoi problem using RMI client/server technique.

   (b) Write a java program to display the users and their IP address logged into the network.

---------- Cut here ----------

5. (a) Write a java program to add two numbers using RMI client/server technique.

   (b) Write a java program set up a chatting window server and client.

---------- Cut here ----------

6. (a) Write a Java program to subtract two numbers using RMI client/server technique.

   (b) Write a java program for bi-directional message communication between client and server.

   (Example : Client sends the message "Hi from client" and the server response will be "Hello" from server)

---------- Cut here ----------

7. (a) Write a java program to set up a time-server and time-client.

   (b) Write a java program to find the square of a number using RMI client/server.

8. (a) Write a java program to find the reverse of a given string using RMI client/server technique.

   (b) Write a java program to implement the client-server communication using TCP.

---------- Cut here ----------

9. (a) Write a java program to check the given string is PALINDROME or NOT using RMI client/server technique.

   (b) Write a java program to ENCRYPT and DECRYPT the word at client and server respectively.

---------- Cut here ----------

10. (a) Write a java program to implement the client-server communication using UDP.

    (b) Write a java program to generate Fibonacci series of given number using RMI client/server technique.

DE–8823 **56**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Fifth Semester

LAB IX — VISUAL C++

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system.

Each question carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write visual C++ win 32 application program using MFC that creates a new font.

   (b) Write visual C++ win 32 application program using MFC that displays a message “Hello”.

2. (a) Write visual C++ win 32 application program "Good Morning!" whenever the user clicks the mouse button on the client area.

   (b) Write visual C++ win 32 application program using MFC that allows the user to draw pictures with the help of mouse as a free hand drawing tool.

---------- Cut here ----------

3. (a) Write visual C++ win 32 application program using MFC that creates a list box and displays name of the states in India.

   (b) Write visual C++ win 32 application program using MFC that displays line, rectangle, rounded rectangle, ellipse and polygon filled colors.

---------- Cut here ----------

4. (a) Write visual C++ win 32 application program that fills the background of the client area with a bitmap.

   (b) Write visual C++ win 32 application program using MFC that displays a menu. Choose the menu items using keyboard accelerator keys and display appropriate messages for the selected command, in message box.

---------- Cut here ----------

5. (a) Write visual C++ win 32 application program using MFC that displays the status of ALT, CTRL, SHIFT, NUMLOCK and SCROLL LOCK Keys.

   (b) Write visual C++ win 32 application program using MFC that displays current mouse coordinates in status bar.

6. (a) Write visual C++ win 32 application program using MFC that creates two push buttons OK and CANCEL on the client area. Buttons should respond to user click over them and display appropriate message.

   (b) Write visual C++ win 32 application program using MFC that draws straight lines when the user clicks left mouse button on start point, then drag and release the left mouse button at end point.

---------- Cut here ----------

7. (a) Write visual C++ win 32 application program using MFC that creates a list box and displays name of the district in Tamilnadu state.

   (b) Write visual C++ win 32 application program using MFC that displays a message "HOW ARE YOU?"

---------- Cut here ----------

8. (a) Write visual C++ win 32 application program using MFC that creates a list box and displays name of the staff members in your department.

   (b) Write visual C++ win 32 application program using MFC that displays a message "Good Afternoon".

---------- Cut here ----------

9. (a) Write visual C++ win 32 application program using MFC that creates a list box and displays name of the various courses available in your college.

   (b) Write visual C++ win 32 application program using MFC that displays a message "Good Evening".

10. (a) Write visual C++ win 32 application program using MFC that creates a list box and displays name of your family members.

(b) Write visual C++ win 32 application program using MFC that displays a message "God Bless You".

DE–8824 **57**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Fifth Semester

LAB X – COMPILER DESIGN

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by LOT system.

Break-up of marks :
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. Write a C program to accept a string and check whether if is an operator, grammar or not

   (a) $S \to aBcD; S \to aBdC$

   (b) $E \to E + E / E * E / (E) / id$

   (c) $E \to EOP\ EE \to (E); E \to id, OP \to + \backslash - \backslash * \backslash /$.

---------- Cut here ----------

2. Write a C program to translate the expression $(a+b)*(b-c)/(a+b)$ into postfix notation and three – address code, quadruples, triples and indirect triples.

3. Construct a C program to accept a string and check whether it satisfies the following production rule (by using operator precedence grammar)

$S \rightarrow AbB$

$A \rightarrow aBb$

$B \rightarrow a / Bba$.

---------- Cut here ----------

4. Write a C program to construct of an canonical LR parsing table for the following grammar

$S \rightarrow CC$

$C \rightarrow CC / d$.

---------- Cut here ----------

5. Write a C program to accept a string and check whether it belongs to ambiguous grammar or not by using following production rule.

$E \rightarrow E + E / E - E / E * E / E / E / id / const / (E)$.

---------- Cut here ----------

6. Write a C program to construct of an SLR parsing table for the following grammar.

$E \rightarrow E + T / T$

$T \rightarrow T * F / F$

$F \rightarrow (E) / id$.

7. Write a C program to construct the left - most derivation and right – most derivation for the following grammar

$S \rightarrow a / \lambda / (T)$

$T \rightarrow ,T,S / S$

(a) (a, (a,a))

(b) To find the LEADING and TRAILING sets.

---------- Cut here ----------

8. Write a C program to construct predictive parsing table for the following production rule.

$S \rightarrow iCtSSi / a$

$S_1 \rightarrow eS / E(E \text{ is empty})$

$C \rightarrow b$.

---------- Cut here ----------

9. Write a C program for the following :

(a) Remove spaces

(b) Comment lines

(c) New lines and

(d) produce line numbers.

---------- Cut here ----------

10. Write a C program for the following :

Recognize identifier in the given source code.

DE–8825 **65**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2014.

Sixth Semester

Computer Applications

LAB XI — MULTIMEDIA APPLICATIONS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the ONE question for each candidate by lot system. Each subdivision carries 50 marks.

1. (a) Write a C++ program for setting up a pattern for multi—grey—level images or bilevel display using limbs algorithm with the following 2×2 matrix.

0 2

3 1

Also print the dot patterns produced on a 16 grey-level system for a pixel.

(b) Design a birth day greeting page including graphic features in Flash.

2. (a) Write a C++ program to create a second image array model from the given image using grey-level transformation bunching technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

Also draw histograms for the first and second images.

(b) Design an animated cartoon picture.

---------- Cut here ----------

3. (a) Write a C++ program for creating an image from the given image using automatic selection of grey level For splitting technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

(b) Prepare a multimedia presentation shows the features of a software package developed by you.

---------- Cut here ----------

4. (a) Write a C++ program to find the best gery level for splitting the following image.

4	5	1	7	4	2	6	3	4	2	7	4	6	7
3	7	8	2	7	6	5	4	3	3	7	2	4	3
4	5	9	9	9	7	8	7	6	5	5	4	1	2
0	0	0	3	4	2	2	3	1	1	4	6	6	5
7	3	2	4	5	1	6	2	3	4	5	6	7	9

Using automatic select ion for splitting technique.

(b) Prepare a multimedia presentation showing the features of the courses offered in a College.

5. (a) Write a C++ program for convolution using given template.

Image					Template	
1	1	3	3	4	1	0
1	1	4	4	3	0	1
2	1	3	3	3		
1	1	1	4	4		

(b) Write a C++ program that smoothens an image in the transform domain.

---------- Cut here ----------

6. (a) Write a C++ program to compress the image compression using contour coding technique.

Given image:

1	1	1	1	1	1	1	1	2	2	2	2	2	2	2
1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	2	2	2	2

(b) Write a C++ program for grey level transformation techniques.

(i) Thresholding

(ii) Bunching.

---------- Cut here ----------

7. (a) Write a C++ program to find the edges of the following image using Sobel edge detection procedure.

(b) Write a C++ program for 2 dimensional image animation using computer graphics.

8. (a) Write a C++ program to perform image subtraction operation.

   (b) Do the image animation using Flash.

---------- Cut here ----------

9. (a) Write a C++ program to compress an image using contour coding technique.

   (b) Design an invitation for your college day function including various graphic features in flash.

---------- Cut here ----------

10. (a) Design a sales advertisement for a product using Flash.

    (b) Write a C++ program to enhance an image using median filter.